



Answers & Solutions:

1. **A.** $\frac{Z_{out}}{1+A\beta}$

2. **A.** three

3. **B.** open loop

4. **C.** from the center of split capacitors

5. **A.** positive

6. **B.** reduce distortion

7. **A.** 0.033

Solution:

$$A_f = \frac{A}{1+A\beta}$$

$$12 = \frac{20}{1+(20)\beta}$$

$$\beta = 0.033 \rightarrow \text{Ans.}$$

8. **B.** is increased

9. **A.** 0.05

Solution:

$$\%d_f = \frac{\%d_i}{1+A\beta}$$

$$0.1\% = \frac{6\%}{1+\left(\frac{25\text{ V}}{20\text{ mV}}\right)\beta}$$

$$\beta = 0.05 \rightarrow \text{Ans.}$$

10. **C.** $A\beta \gg 1$

11. **A.** hand capacitance

12. **B.** undamped

13. **A.** hand capacitance

14. **A.** three

15. **B.** d.c. power into a.c. power

16. **B.** output current

17. **C.** is decreased

18. **A.** -3.85

Solution:

$$A_f = \frac{A}{1+A\beta}$$

$$A_f = \frac{-100}{1+(-100)(-0.25)}$$

$$A_f = -3.85 \rightarrow \text{Ans.}$$

19. **D.** 100 %

20. **D.** Wien bridge oscillator

21. **A.** is increased

22. **C.** both positive and negative

23. **C.** sacrifice factor

24. **A.** radio receivers

25. **C.** transistor

26. **C.** 126.5 pF, 31.6 pF

Solution:

$$\beta = \frac{C_1}{C_2} = 0.25$$

$$C_2 = 4C_1$$

$$f_r = \frac{1}{2\pi\sqrt{LC_T}}$$

$$1\text{ MHz} = \frac{1}{2\pi\sqrt{(1\text{ mH})C_T}}$$

$$C_T = 25.3\text{ pF}$$

$$C_T = \frac{C_1 C_2}{C_1 + C_2} = 25.3\text{ pF}$$

$$25.3\text{ pF} = \frac{C_1(4C_1)}{C_1 + 4C_1}$$

$$C_1 = 31.6\text{ pF} \rightarrow \text{Ans.}$$

$$C_2 = 4C_1 = 126.5\text{ pF} \rightarrow \text{Ans.}$$

27. **A.** decreased 2 times

28. **A.** inversely proportional to square root of

29. **A.** $\frac{D}{1+A\beta}$

30. **B.** minimum

31. **B.** R and C

32. **C.** 390 Ω

Solution:

$$Z_1 = R_1(1+AB)$$

$$= 15\text{ k}\Omega[1+(-100)(-0.25)]$$

$$Z_1 = 390\text{ k}\Omega \rightarrow \text{Ans.}$$

33. **B.** three RC sections

34. **B.** $\frac{1}{2\pi RC}$

35. **B.** $Z_{in}(1+A\beta)$

36. **A.** is independent of

37. **A.** $\frac{1}{2\pi\sqrt{LC}}$

38. **C.** 0.01

Solution:

$$\beta = \frac{100\text{ mV}}{10\text{ V}}$$

$$\beta = 0.01 \rightarrow \text{Ans.}$$

39. **A.** 769 Ω

Solution:

$$Z_o = \frac{R_o}{(1+AB)} = \frac{20\text{ k}\Omega}{1+(-100)(-25)}$$

$$Z_o = 769\text{ }\Omega \rightarrow \text{Ans.}$$

40. **B.** is reduced

41. **D.** $\frac{1}{A_f} - \frac{1}{A}$

42. **A.** decreased 2 times

43. **B.** it has superior electrical

44. **C.** 100



Solution:

$$A_f \approx \frac{1}{\beta} \approx \frac{1}{0.01}$$

$$A_f \approx 100 \rightarrow \text{Ans.}$$

45. **D. more than 10,000**

46. **C.** $\frac{1}{2\pi RC\sqrt{6}}$

47. **A. a voltage developed because of mechanical stress**

48. **A. resistive**

49. **D. klystron**

50. **A. less than 1**

END

